

IRR: Definition, Biases, and Limitations

Session 6 • The dominant private market return metric and its problems

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What we'll cover today

1

IRR: formal definition

What the math actually says

2

Why IRR is dominant in PE

Industry conventions

3

Bias 1: Cash flow timing

Reinvestment assumption

4

Bias 2: Reporting choices

Cumulative vs. since-inception

5

Bias 3: Multi-fund aggregation

Pooled vs. arithmetic IRR

IRR: the formal definition

Internal Rate of Return — the discount rate that sets NPV to zero:

$$\sum_{t=0}^T \frac{CF_t}{(1 + IRR)^t} = 0$$

Where:

CF_0	Initial capital call (negative)
$CF_t, t \in (0, T)$	Periodic cash flows (calls negative, distributions positive)
CF_T	Final cash flow (distributions + terminal NAV)
IRR	The discount rate that solves the equation
T	Time horizon (typically 10+ years for closed-end PE)

IRR worked example: a typical PE fund

Hypothetical 10-yr PE fund: \$100M commitment, calls drawn 2017-2020, distributions 2021-2027.

Year	Calls	Distributions	Net Cash Flow	Cumulative
2017	-\$25M	\$0	-\$25M	-\$25M
2018	-\$30M	\$0	-\$30M	-\$55M
2019	-\$25M	\$5M	-\$20M	-\$75M
2020	-\$20M	\$10M	-\$10M	-\$85M
2021	\$0	\$15M	+\$15M	-\$70M
2022	\$0	\$30M	+\$30M	-\$40M
2023	\$0	\$45M	+\$45M	+\$5M
2024	\$0	\$60M	+\$60M	+\$65M
2025	\$0	\$50M	+\$50M	+\$115M
2026	\$0	\$25M	+\$25M	+\$140M
2027	\$0	\$10M	+\$10M	+\$150M

Why IRR dominates private market reporting

REASON 1

Captures both cash and timing

Unlike simple multiples (TVPI), IRR incorporates time-value of money. A 2.5× over 5 years is different from 2.5× over 10 years.

→ Mathematically: $IRR \neq f(TVPI)$ alone

REASON 2

Familiar from corporate finance

Standard CFA / MBA training. Investment committees understand 'IRR' even when they don't understand PE-specific issues. Lowers communication friction.

→ Network effect: everyone uses it

REASON 3

Industry GIPS standard

Global Investment Performance Standards (GIPS) mandate IRR reporting. Regulatory and institutional infrastructure built around it.

→ Sunk-cost lock-in: changing is expensive

None of these are mathematically defensible reasons. They are sociological reasons.

Bias 1: The reinvestment assumption

IRR implicitly assumes distributions are reinvested at the IRR rate itself.

If your fund's IRR = 18%, IRR-math says: 'Treat all distributions as if you could redeploy them at 18% until fund maturity.'

But realistically:

- ▶ **LP investing in PE-of-PE funds** Redeploys early distributions to other PE funds at similar IRRs — assumption ~holds
- ▶ **LP in pension fund context** Redeploys to public markets at ~7-8% — assumption far overstated
- ▶ **LP in continuation fund vehicle** May NOT redeploy at all (returned to general portfolio) — assumption invalid
- ▶ **LP in late-vintage PE position** Cannot redeploy at 18% — distributions go to lower-return alternatives

Modified IRR (MIRR) corrects this by specifying explicit reinvestment rate. Rarely used in industry.

Bias 2 & 3: Reporting choices and aggregation

BIAS 2

Cumulative vs. since-inception

Two ways to report:

Cumulative IRR (since fund inception)

Best of both worlds for early-vintage funds reporting on a strong run.

Net (after fees) vs. gross

Some GP marketing materials report gross. Investor cost of capital is net.

Rolling window vs. since inception

Rolling 5-yr IRR can look very different from since-inception IRR.

BIAS 3

Multi-fund aggregation

Two aggregation methods:

Pooled IRR

Combine cash flows across all funds, compute one IRR. Gives weight by dollar volume.

Average IRR (arithmetic mean)

Compute IRR per fund, take simple mean. Weights all funds equally regardless of size.

Typical gap:

Pooled IRR may exceed average IRR by 2–5 percentage points (size and reinvestment effects).

Session 6 summary

What we accomplished today

- 1 IRR is the dominant return metric in private markets — formal definition is $NPV = 0$ discount rate
- 2 Three structural reasons IRR dominates: math, familiarity, GIPS standard
- 3 Three structural biases: reinvestment assumption, reporting choice asymmetry, aggregation method
- 4 These biases collectively can move reported IRR by 200-500 bps — material at the asset class level

Next session

Session 7: PME, LA-IRR, LA-PME — the metrics that compete with IRR